

H5 Integrating fractions A level only

Name: _____

Class: _____

Date: _____

Time: **145 minutes**

Marks: **119 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

Find the value of $\int_1^2 \frac{6x+1}{6x^2-7x+2} dx$, expressing your answer in the form $m \ln 2 + n \ln 3$, where m and n are integers.

(Total 8 marks)

Q2.

(a) (i) Express $\frac{5x-6}{x(x-3)}$ in the form $\frac{A}{x} + \frac{B}{x-3}$.

(2)

(ii) Find $\int \frac{5x-6}{x(x-3)} dx$.

(2)

(b) (i) Given that

$$4x^3 + 5x - 2 = (2x + 1)(2x^2 + px + q) + r$$

find the values of the constants p , q and r .

(4)

(ii) Find $\int \frac{4x^3 + 5x - 2}{2x + 1} dx$.

(3)

(Total 11 marks)

Q3.

(a) Find $\int \frac{1}{3+2x} dx$.

(2)

(b) By using integration by parts, find $\int x \sin \frac{x}{2} dx$.

(4)

(Total 6 marks)

Q4.

(a) Express $\frac{1}{(3-2x)(1-x)^2}$ in the form $\frac{A}{3-2x} + \frac{B}{1-x} + \frac{C}{(1-x)^2}$.

(4)

(b) Solve the differential equation

$$\frac{dy}{dx} = \frac{2\sqrt{y}}{(3-2x)(1-x)^2}$$

where $y = 0$ when $x = 0$, expressing your answer in the form

$$y^p = q \ln[f(x)] + \frac{x}{1-x}$$

where p and q are constants.

(9)

(Total 13 marks)

Q5.

The expression $\frac{10x^2 + 8}{(x+1)(5x-1)}$ can be written in the form $2 + \frac{A}{x+1} + \frac{B}{5x-1}$, where A and B are constants.

(a) Find the values of A and B .

(4)

(b) Hence find $\int \frac{10x^2 + 8}{(x+1)(5x-1)} dx$.

(4)

(Total 8 marks)

Q6.

(a) (i) Express $\frac{7x-3}{(x+1)(3x-2)}$ in the form $\frac{A}{x+1} + \frac{B}{3x-2}$.

(3)

(ii) Hence find $\int \frac{7x-3}{(x+1)(3x-2)} dx$.

(2)

(b) Express $\frac{6x^2 + x + 2}{2x^2 - x + 1}$ in the form $P + \frac{Qx+R}{2x^2 - x + 1}$.

(3)

(Total 8 marks)

Q7.

(a) (i) Express $\frac{2x+7}{x+2}$ in the form $A + \frac{B}{x+2}$, where A and B are integers.

(2)

(ii) Hence find $\int \frac{2x+7}{x+2} dx$.

(2)

(b) (i) Express $\frac{28+4x^2}{(1+3x)(5-x)^2}$ in the form $\frac{P}{1+3x} + \frac{Q}{5-x} + \frac{R}{(5-x)^2}$, where P , Q and R are constants.

(5)

(ii) Hence find $\int \frac{28+4x^2}{(1+3x)(5-x)^2} dx$.

(4)

(Total 13 marks)

Q8.

(a) (i) Given that $y = 2x^2 - 8x + 3$, find $\frac{dy}{dx}$.

(1)

(ii) Hence, or otherwise, find

$$\int_4^6 \frac{x-2}{2x^2-8x+3} dx$$

giving your answer in the form $k \ln 3$, where k is a rational number.

(4)

(b) Use the substitution $u = 3x - 1$ to find $\int_4^6 \frac{x-2}{2x^2-8x+3} dx$, giving your answer in terms of x .

(4)

(Total 9 marks)

Q9.

(a) Given that $\frac{3}{9-x^2}$ can be expressed in the form $k\left(\frac{1}{3+x} + \frac{1}{3-x}\right)$, find the value of the rational number k .

(2)

(b) Show that $\int_1^2 \frac{3}{9-x^2} dx = \frac{1}{2} \ln\left(\frac{a}{b}\right)$, where a and b are integers.

(3)

(Total 5 marks)

Q10.

- (a) Express $\frac{2}{x^2 - 1}$ in the form $\frac{A}{x-1} + \frac{B}{x+1}$. (3)

- (b) Hence find $\int \frac{2}{x^2 - 1} dx$. (2)

- (c) Solve the differential equation $\frac{dy}{dx} = \frac{2y}{3(x^2 - 1)}$, given that $y = 1$ when $x = 3$.

Show that the solution can be written as $y^3 = \frac{2(x-1)}{x+1}$. (5)

(Total 10 marks)

Q11.

- (a) (i) Express $\frac{3x-5}{x-3}$ in the form $A + \frac{B}{x-3}$, where A and B are integers. (2)

- (ii) Hence find $\int \frac{3x-5}{x-3} dx$. (2)

- (b) (i) Express $\frac{6x-5}{4x^2-25}$ in the form $\frac{P}{2x+5} + \frac{Q}{2x-5}$, where P and Q are integers. (3)

- (ii) Hence find $\int \frac{6x-5}{4x^2-25} dx$. (3)
- (Total 10 marks)

Q12.

- (a) (i) Given that $f(x) = x^4 + 2x$, find $f'(x)$. (1)

- (ii) Hence, or otherwise, find $\int \frac{2x^3 + 1}{x^4 + 2x} dx$. (2)

- (b) (i) Use the substitution $u = 2x + 1$ to show that

$$\int x\sqrt{2x+1} \, dx = \frac{1}{4} \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du$$

(3)

- (ii) Hence show that $\int_0^4 x\sqrt{2x+1} \, dx = 19.9$ correct to three significant figures.

(4)

(Total 10 marks)

Q13.

- (a) Given that $\frac{9x^2 - 6x + 5}{(3x-1)(x-1)}$ can be written in the form $3 + \frac{A}{3x-1} + \frac{B}{x-1}$, where A and B are integers, find the values of A and B .

(4)

- (b) Hence, or otherwise, find $\int \frac{9x^2 - 6x + 5}{(3x-1)(x-1)} \, dx$.

(4)

(Total 8 marks)